Resident Solution Architect AI-Driven Infrastructure

HashiCorp Professional Services

THE CHALLENGE

AI for Infrastructure is High-Reward, High-Risk

Enterprises need expert guidance to capture value while managing risk

Security Risks

Overprivileged agents risk infrastructure destruction, secret leakage, and policy violations

Delivery Bottlenecks

Platform teams are capacity-constrained — module demand outpaces delivery by weeks

Workflow Immaturity

Traditional IaC workflows weren't designed for AI velocity — controls can't keep pace

Skill Gaps

App teams lack infrastructure expertise; platform teams lack AI workflow experience

A Resident Solution Architect helps establish the mature practices and strong controls required for safe AI adoption

RESIDENT SOLUTION ARCHITECT

What Your Dedicated Expert Delivers

Embedded with your team, bridging the gap between AI capability and enterprise readiness

Guardrails & Controls

- RBAC and agent isolation patterns
- Secrets management with Vault Radar
- Policy-as-code enforcement
- Human-in-loop gated approvals

Validated Workflows

- Spec-driven development methodology
- Consumer composition from registry
- Module authoring acceleration
- Provider lifecycle development

Accelerated Outcomes

- Module delivery: weeks to hours
- Provider development: months to days
- Composition: hours to minutes
- Team upskilling via paired sessions

PROVEN RESULTS

Validated Use Cases Across Three Personas

Your RSA guides implementation of battle-tested workflows with measurable acceleration

APPLICATION TEAM

Consumer Workflow

Deploy and manage infrastructure using natural language from approved private registry modules

- Self-service provisioning via AI
- Guardrailed module consumption
- Reduced ticket volume to platform team

PLATFORM TEAM

Module Authoring

Generate compliant, tested Terraform modules with spec-driven development and automated validation

- AI-assisted module scaffolding
- Automated testing and compliance checks
- Consistent module standards at scale

ECOSYSTEM

Provider Lifecycle

Accelerate Terraform provider development with TDD, API validation, and SDK migration support

- Faster provider development cycle
- Automated SDK migration support
- Validated API coverage

ENGAGEMENT TIMELINE

Resident Solution Architect: 3–6 Months

A phased engagement that builds capability progressively — from assessment to autonomous operation

1 Assess & Establish

MONTH 1

- IaC maturity assessment
- Guardrails and governance setup
- DevContainer and sandbox config
- First workflow implementation

2 Enable & Accelerate

MONTHS 2 – 4

- Expand to additional use cases
- Team enablement workshops
- Spec-driven methodology adoption
- CI/CD and policy integration

3 Scale & Handoff

MONTHS 5 – 6

- Cross-team adoption patterns
- Document runbooks and standards
- Measure outcomes against KPIs
- Transition to self-sufficient ops

Dedicated RSA: Assigned and embedded with your team for the full engagement. Candid feedback, hands-on delivery, and a clear path to scale.

EXPECTED OUTCOMES

What You Walk Away With

Measurable outcomes delivered across governance, velocity, and team capability

Production-Ready Guardrails

RBAC, policy-as-code, secrets management, and agent isolation configured and validated

Implemented Use Case

At least one AI-powered IaC workflow deployed and operational with your team

Documented Standards

Runbooks, patterns, and best practices codified for repeatable adoption

Enabled Teams

Your engineers trained on spec-driven development and agentic workflows

Speed

Accelerate module delivery and reduce provisioning lead times from days to minutes

Risk

Eliminate misconfigurations before they reach production with automated policy enforcement

Cost

Reduce manual review overhead and right-size infrastructure with AI-driven optimization

Ready to start? Partner with HashiCorp to shape your AI-driven IaC roadmap and implement your first use case with dedicated expert guidance.

STRATEGIC IMPACT

How the Resident Solution Architect Creates Compounding Value

Each capability unlocks the next — building momentum across your organization

01 Establish

Guardrails & Controls

- RBAC and agent isolation
- Secrets management
- Policy-as-code enforcement
- Human-in-loop approvals

Zero unreviewed changes reach production

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02 Enable

Self-Service Infra

- Consumer workflow patterns
- Natural language provisioning
- Registry-backed modules
- Guardrailed consumption

Teams provision in minutes, not days

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03 Accelerate

Pattern Authoring

- Spec-driven development
- AI-assisted module creation
- Automated test generation
- Cross-team module sharing

Module delivery: weeks → hours

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04 Scale

Organization-Wide

- Cross-team adoption
- Consistent standards everywhere
- Self-sufficient operations
- Measurable ROI

Every team benefits from every pattern

★ **Compounding returns** — each stage builds on the last, creating a flywheel of capability that accelerates over time.

PRE-REQUISITES

Enterprise Readiness for AI-Driven IaC

Key foundations your organization should have in place — or that your RSA will help establish

IaC Maturity

- Git-based Terraform workflow
- Private module registry
- CI/CD for plan & apply
- Documented standards

Security & Governance

- RBAC via HCP Terraform
- Secrets mgmt (Vault etc.)
- Policy-as-code framework
- Agent isolation capability

Verification

- Testing practices for IaC
- Dev/sandbox environments
- Observability & audit logging
- Stakeholder alignment

AI & Tooling

REQUIRED

- Frontier model access (API)
- AI coding agent with skills
- Agentic workflow support
- Prompt & context patterns

Don't have all of these in place? That's exactly what the RSA engagement addresses — we meet you where you are and build the foundations together.

Thank You

Let's build the future of AI-driven infrastructure together